

Linear Algebra

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W3 L1 Four Fundamental Subspaces of matrix A

① Column Space $C(A)$:

$$A = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{bmatrix}$$

$$C(A) = \text{Span}(v_1, v_2, \dots, v_n) = \left\{ \text{linear combo of vectors } v_1, \dots, v_n \right\}$$

Solving $Ax = b$:

Q → For what b does $Ax = b$ have a solution?

Ans → $b \in C(A)$

Let's look at an example:-

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

$Ax = b$ is solvable for $b \in C(A)$

Does every vector in $\mathbb{R}^4$ belong to $C(A)$?

Ans Some vectors in $\mathbb{R}^4$ are not in $C(A)$, because $Ax = b$ is "4 eq. in 3 unknowns"

For eg. above, $\text{col } 3 = \text{col } 1 + \text{col } 2$

So, $C(A) = \text{span}(\text{col}(1), \text{col}(2))$

⇒ $C(A)$ is a two-dimensional subspace of $\mathbb{R}^4$

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$\Rightarrow C(A)$ is a two-dimensional subspace of $\mathbb{R}^4$

② Null Space $N(A)$

$$N(A) = \{x \mid Ax=0\} \quad \text{Why is } N(A) \text{ a subspace?}$$

$$x_1, x_2 \in N(A)$$

$$x \in N(A), \alpha \text{ is a scalar}$$

$$\text{Then, } Ax_1=0, Ax_2=0$$

$$A(\alpha x) = \alpha Ax = 0$$

$$\Rightarrow A(x_1+x_2)=0$$

$$\Rightarrow x_1+x_2 \in N(A)$$

$$\text{So, } \alpha x \in N(A)$$

Hence $N(A)$ is a subspace.

eg. $\rightarrow$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

Find x s.t. $Ax=0$

A linear combo of the columns of A should result in zero vector.

$$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \in N(A)$$

So, $N(A)$ is a line in $\mathbb{R}^3$

SIMP

Remark - If A is invertible, then $N(A)$ has "zero" ^{vector} only, and $C(A)$ is the whole space.

In this case, $Ax=b$ has a unique solⁿ $x=A^{-1}b$

Else, $N(A)$ has $x_m \neq 0$ and $Ax=b$ solutions are of the form

$$x = x_p + x_m$$

$$Ax_p = b$$

$$Ax_m = 0$$

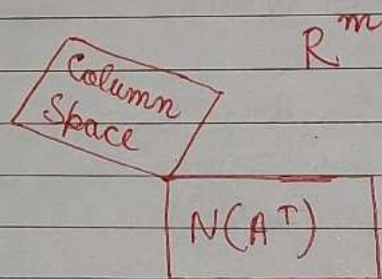
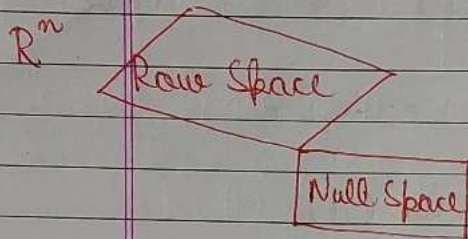
A is $m \times n$

EXTRA

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- 1) Column Space $C(A)$ in $\mathbb{R}^m$
- 2) Null Space $N(A)$ in $\mathbb{R}^n$
- 3) Row Space $C(A^T)$ in $\mathbb{R}^n$
- 4) Left Null Space $N(A^T)$ in $\mathbb{R}^m$



Q- How to produce its basis and Dimension?

Ans $\rightarrow$ Dim. of Col. Space is rank r

Basis is Pivot Columns

Ans $\rightarrow$ Dim. of Row Space is r

$C(A)$	$N(A)$
Basis is Pivot Columns.	Basis is special sol ⁿ
No. of Pivot Variable (r)	$(n-r)$ No. of free variables.

P.T.O

Remark 2:- Can use Gaussian elimination to find the Null Space of a matrix A i.e., solve $Ax=0$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

Gaussian
Elimination

$$\begin{bmatrix} \textcircled{1} & 2 & 2 & 2 \\ 0 & 0 & \textcircled{2} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivots

(Call this U)

$$Ux=0$$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Col. 1 and Col. 3 $\rightarrow$ Pivot Columns

Col. 2 and Col. 4 $\rightarrow$ Free Variables

Picking Free variables = basis entry

$$x_2=1, x_4=0 \Rightarrow 2x_3+4x_4=0 \Rightarrow x_3=0$$

$$x_1+2x_2+2x_3+2x_4=0 \Rightarrow x_1=-2$$

$$\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\in N(A) \quad \text{Set } x_2=0, x_4=1$$

$$\Rightarrow 2x_3+4x_4=0 \Rightarrow x_3=-2$$

$$x_1+2x_2+2x_3+2x_4=0$$

$$\Rightarrow x_1-4+2=0 \Rightarrow x_1=2$$

$$v = \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\in N(A)$$

$$N(A) = \text{span}(U, v)$$

$$\text{Rank} = 2 \quad (\text{No. of pivot columns})$$

✓ Nullity = No. of free variables. Here it is 2

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$$\text{Rank} = \dim(C(A)), \text{Nullity} = \dim(N(A))$$

If A has n columns, then $\text{rank} + \text{nullity} = n$
 $\text{Rank} = r, \text{nullity}(A) = n - r$

Row Space:- $R(A)$

Column space of $A^T \Rightarrow$ span of rows of A

$$R(A) = C(A^T)$$

Fact:- $\text{Col. Rank} = \dim(C(A)), \text{Row-Rank} = \dim(R(A))$

$\text{Col. Rank} = \text{row rank}$

So far we looked at $C(A), N(A), [R(A) = C(A^T)]$

Left Null Space $N(A^T)$

$$N(A^T) = \{y \mid A^T y = 0\} = \{y \mid y^T A = 0\}$$

For a $m \times n$ matrix A ,

$$\begin{bmatrix} y_1 & \dots & y_m \end{bmatrix} \begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} 0 & \dots & 0 \end{bmatrix}$$

A linear combo. of rows leading to zero vector.

Remark: ① A is a $m \times n$ matrix.

$$\dim(C(A)) + \dim(N(A)) = \text{no. of columns of } A = n$$

$"r" + "n-r" = n$

$$\textcircled{2} \dim(C(A^T)) + \dim(N(A^T)) = \text{no. of rows} = m$$

$$\dim(C(A^T)) = r \Rightarrow r + \dim(N(A^T)) = m$$

$$\Rightarrow \dim(N(A^T)) = m - r$$

eg. $A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$ want $y^T A = 0$

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Ans $\rightarrow [1 \ 1 \ -1] \in N(A^T)$

Ans $\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 4 & 4 \end{bmatrix} \rightarrow$

$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (RREF)

$\uparrow$ Non Pivot

$\Rightarrow [-1 \ -1 \ 1]$

eg 2) $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ $m = n = 2$ $\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ $\uparrow$ P.C. $\uparrow$ N.P.C.

$c(A) = \text{line through } \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ rank $(A) = r = 1$

$R(A) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\dim(R(A)) = r = 1$

$N(A^T) = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$, $\dim$ nullity $= 1$

$N(A^T) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ $\dim$ nullity $(A) = n - r = 1$

(No. of Pivot pts. = Rank)

Q-3) Find basis for each of 4 fundamental subspaces of A

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow C(A)$ is vectors corresponding to pivot columns

$$= \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$\Rightarrow R(A)$ is transpose of non zero rows in RREF

$$= \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$\Rightarrow N(A) \text{ is } Ax=0 \text{ i.e. } \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} =$$

$$x_1 - x_3 = 0$$

$$x_2 + x_3 = 0$$

$$x_3 = x_3$$

$$\Rightarrow \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} x_3$$

$$N(A) = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Or Find Non-Pivot columns and change its sign and complete the diagonal

$$\Rightarrow N(A^T) \text{ is } A^T x = 0 \quad (A^T \text{ in RREF})$$

(

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$$

Lecture 3.1
H W Quest

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$$\begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 6 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{RRE})$$

(Pivot Columns)

$$C(A) = \left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 9 \\ 3 \end{pmatrix} \right\}$$

$$N(A^T) =$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \\ 3 & 9 & 3 \\ 2 & 7 & 4 \end{bmatrix} \quad \text{RRE,}$$

$$R(A) = \left\{ \begin{pmatrix} 1 \\ 3 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 3 & 6 \\ 0 & 3 & 6 \end{bmatrix} \rightarrow$$

$$N(A) = \left\{ \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} +1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} = N(A^T)$$

BQ 3.1 1) $C(A), R(A), N(A)$ of a matrix A .

1) What is true about column space of matrix A ?

It consists of L.C of column vectors of A .
 " is span of the column vectors of A .

2) If x and y belong to null space of A , then L.C of x and y also belong to null space of A .

True! ✓

3) True about Null space of matrix A ?

Those x which satisfy $Ax = 0$

4) A vector x belong to null space of matrix A if:-

$$Ax = 0$$

5) Gaussian elimination changes the null space of matrix.

False! ✓

6) If a $n \times n$ matrix A is invertible then false is:-

True are $\rightarrow C(A) = \mathbb{R}^n$, $N(A)$ only contain 0 vector.
 $\Rightarrow$ False is $Ax = b$ has infinite sol.

7) True about $m \times n$ matrix A ?

$$\dim(C(A)) + \dim(N(A)) = n, \quad \text{Rank}(A) + \text{Nullity}(A) = n$$

$$\dim(C(A^T)) + \dim(N(A^T)) = m$$

8) True about Rank of Matrix?

Dim. of col. space or Dim. of row space

W3L2

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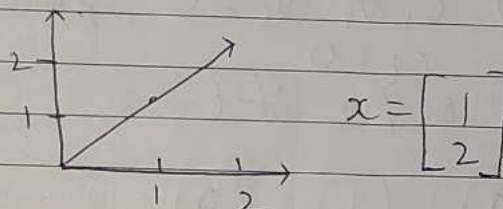
Orthogonal Vectors and Subspaces :-

Least Square :-

Background on orthogonality :-

- We'll look at
- ① Orthogonal vectors, length of vector
 - ② Projections
 - ③ Least Square Regression

① Length of a Vector :-



$$\|x\|^2 = x_1^2 + x_2^2 \quad (\text{Squared length})$$

$$\left\| \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\|^2 = 1^2 + 2^2 = 5$$

In gen., for $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$, $\|x\|^2 = x_1^2 + \dots + x_n^2$

(Length of $x = \sqrt{x^T x}$)

② Orthogonal Vectors :-

→ (Inner or Dot product)

$x \perp y$ if $x^T y = 0$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \quad \text{Then } x^T y = \sum_{i=1}^n x_i y_i$$

From Pythagoras Theorem: $\|x\|^2 + \|y\|^2 = \|x+y\|^2$

$$x^T x + y^T y = (x+y)^T (x+y)$$

$$\Rightarrow x^T x + y^T y = x^T x + y^T y + x^T y + y^T x$$

$$2x^T y = 0 \quad \Rightarrow \quad x^T y = 0 \quad (x \text{ is orthogonal to } y)$$

Remark: 0 is orthogonal to every x . "non trivial set of vectors"

2) If $\{v_1, \dots, v_k\}$ are mutually orthogonal, then $\{v_1, \dots, v_k\}$ is a linearly independent set.

Why? Suppose $c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0$
 $\Rightarrow v_1^T (c_1 v_1 + \dots + c_k v_k) = 0$

$$\Rightarrow c_1 v_1^T v_1 = 0, \text{ (but } \|v_1\| \neq 0 \Rightarrow c_1 = 0)$$

Similarly $c_i = 0 \forall i$

eg: 1) $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

2) $\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$

Orthormal Vectors: $\{u, v\}$ if $u^T v = 0$ and $\|u\| = \|v\| = 1$
 (orthogonal) (unit length)

eg: $\left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}, \left\{ \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} \right\}$

Orthogonal Subspaces: U, V are orthogonal subspaces if

$$x^T y = 0 \quad \forall x \in U, y \in V$$

eg: ① $\{0\}$ $\perp$ subspace

② $U = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}, V = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \quad U \perp V$

$$W = \text{span} \left\{ \begin{pmatrix} 0 \\ 0 \\ -1 \\ 2 \end{pmatrix} \right\}$$

Then $W \perp U, W \perp V$

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Orthogonality w.r.t 4 fundamental subspaces of Matrix A.

Claim: ① $R(A) \perp N(A)$

Proof $x \in N(A)$

$$Ax = 0 \quad \left(A \text{ is } m \times n \right)$$

$$\begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_m \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

The equation above implies $\text{row } 1 \perp x$
 $\text{row } 2 \perp x$
 $\vdots$
 $\text{row } m \perp x$

Thus, any L.C $(C_1 \text{row } 1 + C_2 \text{row } 2 + \dots + C_m \text{row } m) \perp x$
 $\in R(A)$

$$\text{So } R(A) \perp N(A)$$

$$\Leftrightarrow C(A^T) \perp N(A)$$

Claim 2: $C(A) \perp N(A^T)$

eg. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$

Ans $\rightarrow \text{Rank}(A) = \dim(C(A)) = 1$
 $x = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \in N(A)$ since "col 2 = 2 x col 1"

Row space $R(A) = \text{Line through } \begin{bmatrix} 1 & 2 \end{bmatrix}$

Null space $N(A) = \text{Line } \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = 0 \Rightarrow R(A) \perp N(A)$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$$

RREF

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \xrightarrow{\text{P.C.}}$$

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$$\Rightarrow R(A) = \begin{bmatrix} 1 & 2 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \xrightarrow{\text{RREF}}$$

$$C(A) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{N-PC}}$$

$$N(A) = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$\Rightarrow$

$$N(A^T) = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}$$

$$N(A^T) = y_1 + 2y_2 + 3y_3 = 0 \quad \leftarrow \text{is a plane}$$

$$y^T A = 0$$

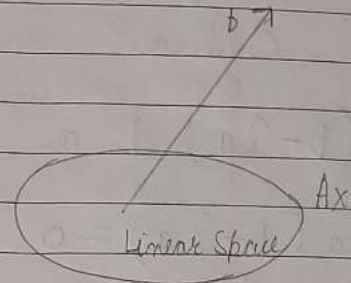
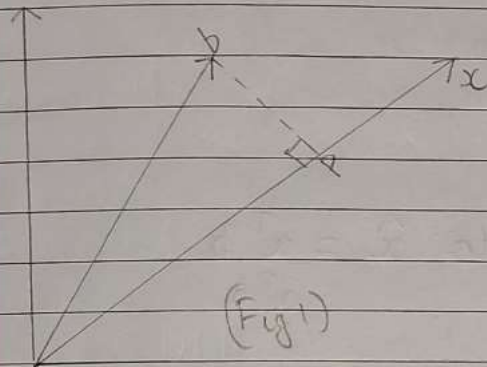
Dimension check:-

$$\dim C(A) + \dim N(A) = 2$$

(1) (1) (No. of columns)

$$\dim C(A^T) + \dim N(A^T) = 3$$

(1) (2) (No. of rows columns)



want to project b onto the line through x , or more generally onto the column space of matrix A .

- ① Why Project?
- ② Given a basis for a subspace S (spanned by columns of A) is there an easy way to calculate projection P of b onto S ?

Ans-1) Suppose we are given $(x_1, b_1), \dots, (x_n, b_n)$
 $2x = b_1$ $x + 2y = 4$
 eg. $3x = b_2$ or $x + 3y = 5$
 $4x = b_3$ $2x + 4y = 6$

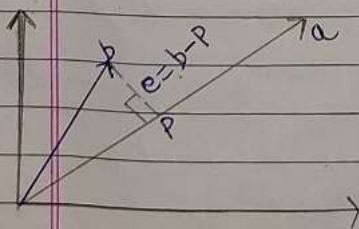
"These are inconsistent" $\Rightarrow$ No solⁿ that satisfies this S.O.E.

Matrix View: $Ax = b$

✓ Inconsistent if $b \notin C(A)$

In such situations it makes sense to project b onto $C(A)$

Projection onto a line:-



$$P = \hat{x}a$$

$$e = b - P = b - \hat{x}a$$

$$e \perp a$$

$$(b - \hat{x}a) \perp a$$

$$a(b - \hat{x}a) = 0 \quad \text{leading to} \quad \hat{x} = \frac{a^T b}{a^T a}$$

$$\Rightarrow P = \hat{x}a = \left(\frac{a^T b}{a^T a} \right) a$$

Cauchy Schwarz Inequality:

$$\|e\|^2 = \|b - P\|^2 \geq 0$$

$$\begin{aligned} \left\| b - \frac{a^T b}{a^T a} a \right\|^2 &= b^T b - 2 \frac{(a^T b)^2}{a^T a} + \left(\frac{a^T b}{a^T a} \right)^2 a^T a \\ &= \frac{(b^T b)(a^T a) - (a^T b)^2}{(a^T a)} \geq 0 \end{aligned}$$

$$\Rightarrow (b^T b)(a^T a) \geq (a^T b)^2$$

$$(or) \quad \boxed{|a^T b| \leq \|a\| \|b\|} \quad |a^T b| \geq -\|a\| \|b\|$$

Projection Matrix

Recall $P = \left(\frac{a^T b}{a^T a} \right) a = \left(\frac{a a^T}{a^T a} \right) b$

Let $P = \frac{a a^T}{a^T a}$ Then, projection of b onto a is $P b$.

To project any vector b , just left multiply by the projection matrix ' P '.

eg. $a = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ Projection Matrix is $P = \frac{a a^T}{a^T a}$ digit $\swarrow$ (Actually Multiply)

Observe that i) P is symmetric

ii) $P^2 = P$ i.e., $P^2 b = P b$

(Idea is: $P b$ is already on line through a . So, another round of projections won't change it)

iii) Column space $C(P) = \text{Line through } a = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$
Rank $r(P) = 1$

iv) $a = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$ $P = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$

(Projection won't change)

AQ 3.3 1) A system of equations $Ax = b$ is inconsistent if:
Ans $b \notin$

2) C.S of projection matrix of vector v is

Line perp to line through v .

3) Consider the following statements regarding a projection matrix P .

- 1) $P^2 = P$ 2) P is symmetric 3) $P^2 \neq P$
 4) P is not symmetric

1 and 2

4) Null space of the projection matrix of vector v is?

Line perp. $\times$ Plane orthogonal to v .

5) Projection Matrix of vector a is given by

$$\frac{aa^T}{a^T a} \quad \frac{aa^T}{\|a\|^2}$$

W3L4 Least Squares and Projections onto a Subspace:-

$$\left. \begin{array}{l} 2x = b_1 \\ 3x = b_2 \\ 4x = b_3 \end{array} \right\} \text{ This system is solvable if } b \text{ is on the line through } \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$$

Suppose we have a vector ' b ' that leads to an "inconsistent" system.

We could pick a subset of equations and solve it exactly. Problem with this approach: Large errors in some inputs and no error in others.

Reasonable Alternative: Minimize Average Error

$$E^2 = (2x - b_1)^2 + (3x - b_2)^2 + (4x - b_3)^2$$

want to minimize the sum of squares E^2

$$\frac{dE^2}{dx} = 0 \quad (\Rightarrow) \quad 2[2(2x-b_1) + 3(3x-b_2) + 4(4x-b_3)] = 0$$

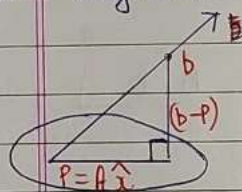
leading to $\hat{x} = \frac{2b_1 + 3b_2 + 4b_3}{2^2 + 3^2 + 4^2} = \frac{a^T b}{a^T a}$ with $a = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$

connects to projections

Bottomline:- Taking derivative and finding the minima turns out to be same as performing a projection

Projection onto a Subspace:-

$Ax=b$, A is $m \times n$, $m > n$
Want: Projection of b onto column space $C(A)$



$S = \text{span}(\text{column of } A)$

Projection of b onto S is $P = A\hat{x}$
Orthogonal Vector $e = b - P = b - A\hat{x}$

Q:-> How to find $\hat{x}$?

Observe: $e \perp$ every vector in $C(A)$

Recall that $C(A) \perp N(A^T)$, $N(A^T)$ is orthogonal complement of $C(A)$, i.e., every vector in $C(A)$ is orthogonal to every vector in $N(A^T)$ & given ~~given~~ vector is in either $C(A)$ or $N(A^T)$.

Where does e belong? $e \in N(A^T) \Rightarrow A^T e = 0$

$$\Rightarrow A^T(b - A\hat{x}) = 0 \quad \text{leading to} \quad \boxed{A^T A \hat{x} = A^T b}$$

eq. to solve to obtain the projection of b onto $C(A)$

(Note:- Even if $Ax=b$ is not solvable, $A^T A \hat{x} = A^T b$ has a solⁿ)

Alternative route to the above eqn :-

$$A = \begin{bmatrix} 1 & | & & | \\ a_1 & a_2 & - & - & a_n \\ | & | & & & | \end{bmatrix}$$

$$\begin{aligned} a_1^T e &= 0 & \Rightarrow & a_1^T (b - A\hat{x}) = 0 & \Rightarrow \\ a_n^T e &= 0 & & a_n^T (b - A\hat{x}) = 0 & \Rightarrow \end{aligned}$$

$$\begin{bmatrix} a_1^T \\ \vdots \\ a_n^T \end{bmatrix} \begin{bmatrix} b - A\hat{x} \end{bmatrix} = 0 \quad \Rightarrow \quad A^T (b - A\hat{x}) = 0$$

leading to $A^T A \hat{x} = A^T b$

Bottomline:-

Solving $A^T A \hat{x} = A^T b$ leads to an $\hat{x}$ that minimizes $\|Ax - b\|_2^2$
(Connection of Projection to least square)

Remark:- (1) Suppose columns of A are linearly independent (l.i.)
Then, $A^T A$ is invertible (why?)
Square, symmetric matrix.

Solving $A^T A \hat{x} = A^T b$ when $A^T A$ is invertible

$$\hat{x} = (A^T A)^{-1} A^T b$$

Projection $P = A \hat{x} = A (A^T A)^{-1} A^T b$

② $b \in C(A)$ i.e., $b = Ax$

$$P = A(A^T A)^{-1} A^T b = A \underbrace{(A^T A)^{-1} A^T A}_I x = Ax = b$$

③ $b \in N(A^T)$

$$P = A(A^T A)^{-1} A^T b = 0 \quad \text{Since } A^T b = 0$$

④ A is square & invertible matrix. $(\Rightarrow) C(A) = \mathbb{R}^m$

$$P = A(A^T A)^{-1} A^T b = A A^{-1} (A^T)^{-1} A^T b = b$$

⑤ A is rank 1, i.e. $A = \begin{bmatrix} 1 \\ \vdots \\ a \end{bmatrix}$

Then, $\hat{x} = \frac{a^T b}{a^T a} \rightarrow$ coincides with what we have derived earlier for projection onto a line.

Projection Matrix $P = A(A^T A)^{-1} A^T$

✓ Symmetric $P^T = P$

$$\begin{aligned} (A(A^T A)^{-1} A^T)^T &= (A^T)^T (A^T A)^{-1})^T A^T \\ &= A(A^T A)^{-1} A^T = P \end{aligned}$$

✓ $P^2 = P$

$$P^2 = A(A^T A)^{-1} A^T A(A^T A)^{-1} A^T = A(A^T A)^{-1} A^T = P$$

So, Projection matrix is symmetric and satisfies $P^2 = P$

The converse is also True.

Claim: If $P^2 = P$ and P is symmetric, then P is a projection matrix.

P_b = Projection of b onto the column space P .

$$(b - P_b)^T P_c = b^T (I - P)^T P c = b^T (P - P^2) c = 0 \quad \left(\begin{array}{l} \text{Data} \\ P = P' \end{array} \right)$$

AQ 3.4 1) If a $n \times n$ matrix is invertible then
 $C(A) = R^n$, $R(A) = R^n$, $\text{Rank}(A) = n$

2) Consider $Ax = b$. The projection matrix of A is I if?
 $b \in C(A)$

3) Consider $Ax = b$. The projection matrix of A is 0 matrix if
 $b \in N(A^T)$

4) Consider full. reg. Projection Matrix P .
 $P^T = P$ $P^T = -P$ $P^2 = P$ $P^2 = -P$

W3L5 Example of Least Square

Simple Case: 1-Dim.

Dataset: $(x_1, b_1), \dots, (x_m, b_m)$

$$b_i = \theta x_i + \theta' \quad , \quad i = 1, \dots, m$$

(Linear Fit)

offset

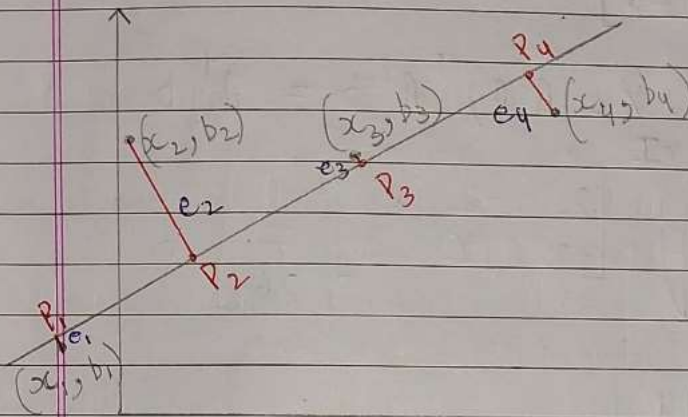
System of eqns (1) is equivalent to

$$\begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} \theta' \\ \theta'' \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \quad A\theta = b \quad \text{where } \theta = \begin{pmatrix} \theta' \\ \theta'' \end{pmatrix}$$

$A\theta = b$ may be inconsistent

Least Squares Approach: minimize $E^2 = \|b - A\theta\|^2$
 $= (b_1 - \theta'x_1 - \theta'')^2 + \dots + (b_m - \theta'x_m - \theta'')^2$

$$(\hat{\theta}', \hat{\theta}'') = \arg \min_{\theta} \|b - A\theta\|^2$$



If the points b_1, b_2, b_3, b_4 lie on a line,
 then $P_1 = b_1, P_2 = b_2, P_3 = b_3, P_4 = b_4$
 $\checkmark$ $E^2 = 0$ and $Ax = b$ can be solved
 (Squared error)

If not then minimize $E^2 = \|A\theta - b\|^2$

eg. $A\theta = b$ $\begin{bmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \theta' \\ \theta'' \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$

1st Check $\rightarrow$ Is $A\theta = b$ consistent? or does $b \in C(A)$?

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 2 \\ 0 & 3 & 5 \end{bmatrix}$$

Inconsistent $A\theta = b$ or $b \notin C(A)$

$$\begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \xleftarrow{\text{Take Projections}} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{bmatrix}$$

Do Least Squares :-

$$A^T A \hat{\theta} = A^T b, \text{ where } \hat{\theta} = \begin{bmatrix} \hat{\theta}' \\ \hat{\theta}'' \end{bmatrix}$$

eg. $\begin{bmatrix} 2 & -2 \\ -2 & 2 \\ 5 & 3 \end{bmatrix} X = \begin{bmatrix} -1 \\ 7 \\ 26 \end{bmatrix}$

$$A^T A x = A^T b \Rightarrow \hat{x} = (A^T A)^{-1} A^T b = A^+ b$$

$$\left(\begin{array}{cc|cc} \xrightarrow{2 \times 3} & \downarrow & & \\ \begin{bmatrix} 2 & -2 & 5 \\ -2 & 2 & 3 \\ 5 & 3 \end{bmatrix} & \begin{bmatrix} 2 & -2 \\ -2 & 2 \\ 5 & 3 \end{bmatrix} & \begin{bmatrix} 2 & -2 & 5 \\ -2 & 2 & 3 \end{bmatrix} & \begin{bmatrix} -1 \\ 7 \\ 26 \end{bmatrix} \\ 2 \times 3 & 3 \times 2 & 2 \times 3 & 3 \times 1 \end{array} \right)^{-1} \mathbf{I}$$

$$\begin{bmatrix} 33 & 7 \\ 7 & 17 \end{bmatrix}^{-1} \begin{bmatrix} 114 \\ 94 \end{bmatrix}$$

$$\begin{bmatrix} 17 & -7 \\ -7 & 33 \end{bmatrix} \begin{bmatrix} 114 \\ 94 \end{bmatrix} = \begin{bmatrix} 1280 \\ 2394 \end{bmatrix}$$

$$\begin{bmatrix} 33 & 7 \\ 7 & 17 \end{bmatrix} \begin{bmatrix} \hat{\theta}' \\ \hat{\theta}'' \end{bmatrix} = \begin{bmatrix} 114 \\ 94 \end{bmatrix}$$

$$33\hat{\theta}' + 7\hat{\theta}'' = 114$$

$$7\hat{\theta}' + 17\hat{\theta}'' = 94$$

$$561\hat{\theta}' + 119\hat{\theta}'' = 1938$$

$$49\hat{\theta}' + 119\hat{\theta}'' = 658$$

$$512\hat{\theta}' = 1280$$

$$\hat{\theta}' = \frac{1280}{512} = 2.5$$

$$\hat{\theta}'' = 4.5$$

Best line (in the "least squares" sense) through the given data is $2.5x + 4.5$

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Ans-2 $A^T A = \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$, $A^T b = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$

$$\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} \hat{\theta}' \\ \hat{\theta}'' \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

$$\hat{\theta}' = 4/7 \quad \hat{\theta}'' = 9/7$$

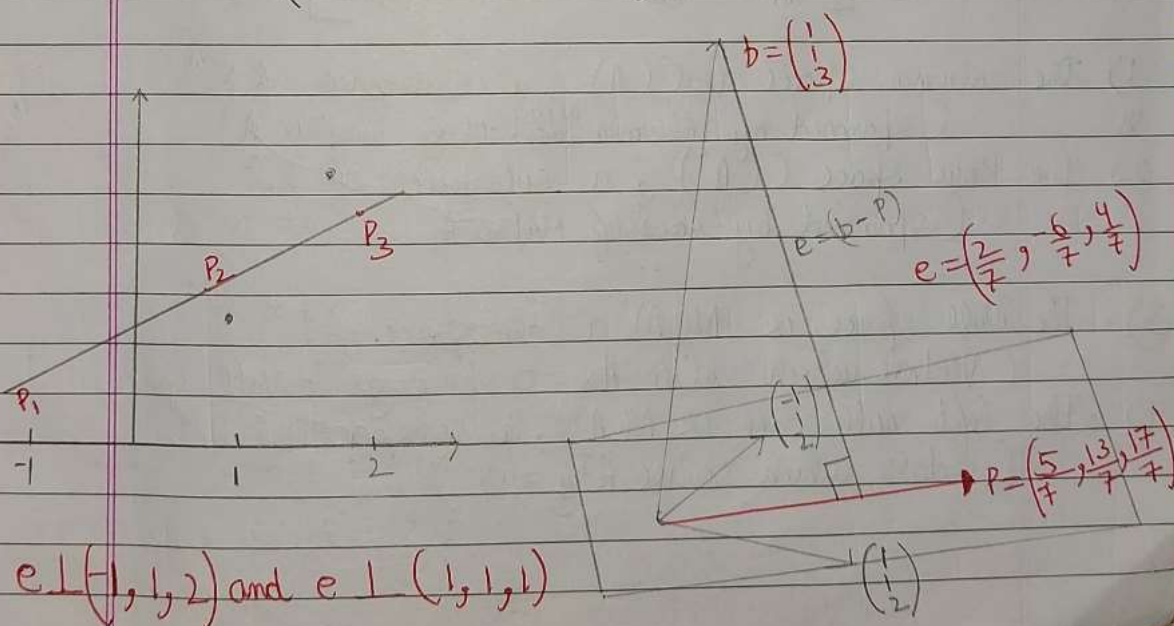
Best line is $(4/7)x + 9/7$

Projections are $P_1 = \frac{4}{7}(-1) + \frac{9}{7} = \frac{5}{7}$, $P_2 = \frac{13}{7}$, $P_3 = \frac{17}{7}$

The original data is not on a line, so $E^2 > 0$

$$\|e\|^2 = E^2 = \|b - A\hat{\theta}\|^2 = \begin{bmatrix} 1 - \left(-\frac{4}{7} + \frac{9}{7}\right) \\ 1 - \left(\frac{4}{7} + \frac{9}{7}\right) \\ 3 - \left(\frac{8}{7} + \frac{9}{7}\right) \end{bmatrix}$$

$$e = \left(\frac{2}{7}, -\frac{6}{7}, \frac{4}{7}\right)$$



$e \perp \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ and $e \perp \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$

D. AQ 3.5

1) If $A\theta = b$ has a solⁿ then,

$$\|A\theta - b\|^2 = 0$$

2) The least squares solⁿ to a system of lin. eqns $A\theta = b$ aims to minimize $\|A\theta - b\|$ 3) The least squares solⁿ to a system of L.E $A\theta = b$ projects b onto $C(A)$ W3 Tutorial 1 Four Fundamental Subspaces :-Suppose A is a $m \times n$ matrix.

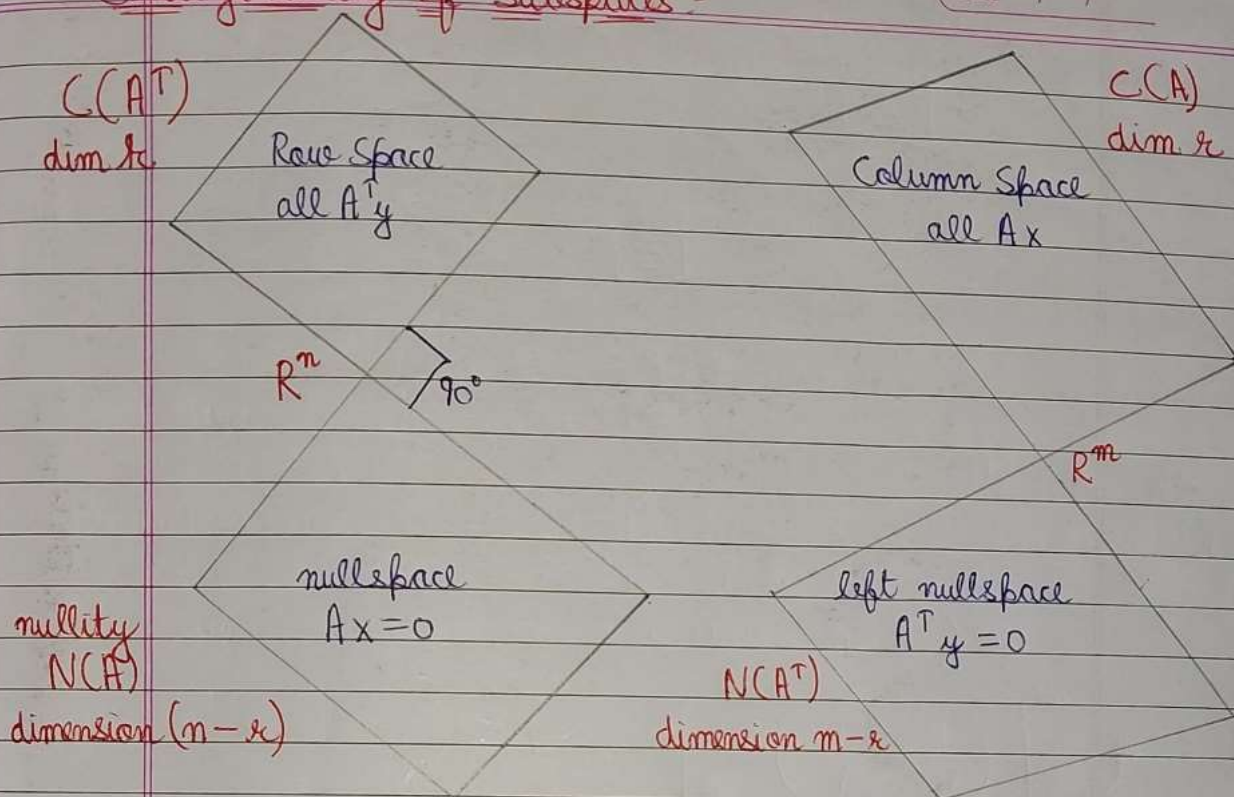
$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- 1) The column space is $C(A)$, a subspace of $\mathbb{R}^m$
(spanned by column ^{vectors} of matrix ~~matrix~~ A)
- 2) The Row Space $C(A^T)$, a subspace of $\mathbb{R}^n$
(spanned by rows of Matrix A)
- 3) The Null space is $N(A)$ a subspace of $\mathbb{R}^n$
(Vectors which solve $Ax = 0$ belongs to Null space)
- 4) The left null space is $N(A^T)$, a subspace of $\mathbb{R}^m$.
(Vectors which solve $A^T y = 0$)

Orthogonality of Subspaces:-

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Every vector in null space is perp. to every row of A .

Dimension of Row space is no. of independent row vectors in the matrix A .

Dim. of Col space // // // // column // in matrix A .

(Both are equal = Rank of Matrix A)

eg:- $A = \begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 2 & 0 & 2 & 4 & -2 \\ 2 & 2 & 4 & 0 & 2 \\ 4 & 4 & 8 & 12 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & -8 & -10 & -12 & -22 \\ 0 & -6 & -8 & -16 & -18 \\ 0 & -12 & -16 & -20 & -32 \end{bmatrix}$

↓

$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & \frac{10}{8} & \frac{12}{8} & \frac{22}{8} \\ 0 & 0 & 1 & -4 & -3 \\ 0 & -12 & -16 & -20 & -32 \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & \frac{10}{8} & \frac{12}{8} & \frac{22}{8} \\ 0 & -6 & -8 & -16 & -18 \\ 0 & -12 & -16 & -20 & -32 \end{bmatrix}$

Notes By Hardik Sharma

If any part is blurred or not understandable
message me on WhatsApp (9877100914)

Tip: Don't directly jump onto the notes firstly
watch the whole video and then read them.